

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 20

Duration : 1 Hour

Scenario Based

Code of Conduct;

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I 192372044 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

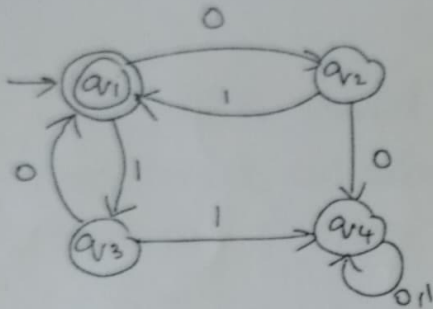
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SIMATS ENGINEERING ASSESSMENT TOOL

1. Given a DFA used for validating a specific PIN pattern bbbc, bca, c, ca, caaaaa minimize the DFA and explain how state reduction improves performance and efficiency in embedded systems

2. Demonstrate how a given finite automaton can be converted into a regular expression



3. Consider the following grammar

$S \rightarrow aSc|T|U|e$

$T \rightarrow bTc|e$

$U \rightarrow aUb|e$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

C Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

4. i) Show that if L_1 and L_2 are regular then $(L_1 + L_2)$ also regular

ii) Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify

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Given a DFA used for validating the strings.

Given strings: bbbb,
bac,
ca,
caaaaa.

The DFA should accept exactly these four strings and reject all others. A Lang-based DFA can be built from the prefixes.

- ϵ
- $b \rightarrow bb \rightarrow bbb \rightarrow bbbb$
- $b \rightarrow ba \rightarrow bac$
- $c \rightarrow ca \rightarrow cca$
- $c \rightarrow ca \rightarrow caa \rightarrow caaa \rightarrow caaaa \rightarrow caaaaa$

Add one dead state for all the invalid transitions.

Minimized DFA:

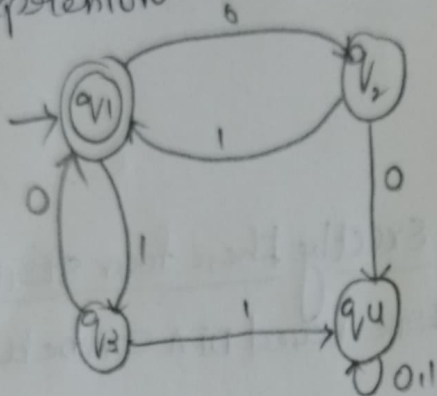
Equivalent dead substates are merged, giving fewer states than raw tree.

Why state reduction improves performance:

- Fewer states $\rightarrow$ Less memory
- Few transition checks
- Faster matching
- Lower power consumption in Embedded Systems.
- Easier testing and maintenance

Conclusion: DFA minimization preserves the Language while reducing hardware/software cost and improving execution efficiency.

2A) Convert the Given finite automaton into a Regular Expression.



Given start state = q_1

Final state = q_4

$\rightarrow q_1 \xrightarrow{0} q_2$ and $q_2 \xrightarrow{1} q_1$

$q_1 \xrightarrow{0} q_3$ and $q_3 \xrightarrow{1} q_1$

$q_2 \xrightarrow{0} q_4$

$q_3 \xrightarrow{1} q_4$

q_4 has loop on (0,1) self-loop

To stay in the accepting state q_4 , the machine can lead only the cycles.

- 01 ($q_1 \rightarrow q_2 \rightarrow q_1$)

- 10 ($q_1 \rightarrow q_3 \rightarrow q_1$)

Any other continuation leads to q_4 .

Hence the Language is any number of repetitions of 01 or 10 including ϵ .

Regular Expression:

$$R.E = (01+10)^*$$

Given Grammar,

$S \rightarrow aSc | \epsilon$

$T \rightarrow bTc | \epsilon$

$U \rightarrow aUble.$

(a) Language Generated

- $S \rightarrow aSc$ adds one a at beginning of c at the end.
- T generates $b^n c^n$.
- U generates $a^m b^m$.

$$\therefore L(S) = \{a^k x c^k \mid k \geq 0, x \in L(TU)\} \cup \{\epsilon\}$$

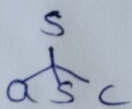
where

$$L(T) = \{b^n c^n \mid n \geq 0\}$$

$$L(U) = \{a^m b^m \mid m \geq 0\}$$

So the strings have balanced out a & c pairs with middle part from TU .

(b) Derivation Tree for $aaabbc$.



$$S \Rightarrow aSc$$

$$\Rightarrow aaSc$$

$$\Rightarrow aaaScc$$

$$\Rightarrow aaATcccc$$

$$\Rightarrow aaAEVccc$$

$$\Rightarrow aaaaubccc$$

$$\Rightarrow aaaaEhccc$$

$$\Rightarrow aaaaabccc$$

The given word $aaabbc$ cannot be generated this grammar.
So No derivation tree exists.

(c) Ambiguity

A grammar is Ambiguous if one string has two distinct parse trees.

- For this grammar, each generated string has a unique decomposition into:

- outer alc pairs from S ,
- abc^n part from T
- an $a^m b^n$ part from U .

Hence the grammar is not ambiguous

(4) (i) If L_1 and L_2 are regular, prove that $(L_1 + L_2)$ is regular

Regular languages are closed under union

Let M_1 and M_2 be DFA's for L_1 and L_2

Construct the product DFA:

- States $Q_1 \times Q_2$
- Start state (q_{01}, q_{02})
- Transition: $\delta((p, q), a) = (\delta_1(p, a), \delta_2(q, a))$
- Final states: all pairs where p is final in M_1 or q is final in M_2 .

This DFA accepts $L_1 \cup L_2$

Therefore $(L_1 + L_2)$ is regular

(ii) Is $L = \{a^p \mid p \text{ is prime}\}$ regular?

Assume L is regular, let pumping length be n .

Choose a prime $p > n$.

Take $s = a^p \in L$

By pumping lemma, write $s = xyz$ with

$$|xy| \leq n$$

$$|y| > 0$$

$$\text{let } |y| = k \geq 1$$

$$\text{pump } i = p + 1$$

$$|xy^{p+1}z| = p + k = p + 1 + (k-1)$$

$\Rightarrow$ the length is composite

$$\text{so } xy^{p+1}z \notin L$$

$\therefore L$ is not regular

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2-3 Marks)	Level 1 – Beginning (0-1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

94/100